energy waveforms. Its rules differ from those of general relativity making the two seem like exact opposites.

Think of it this way: if we hold the theories of general relativity to be true, then it's a fact that satellites revolving around a planet have a decaying orbit and will eventually collide with it over time. By that logic, why don't electrons orbiting the nucleus of an atom also collide? Conversely, quantum mechanics states that we can't ever measure the position and momentum of a particle simultaneously and that all observations are a matter of probability. Under that logical standard, space-time and gravity are fuzzy fluctuations in space. The impasse is exhausting.

To this day, scientists struggle to tie these two concepts together under a unified theory of everything. The search for exotic particles plays a crucial role in determining if these models can ever truly be integrated. The Higgs boson discovery takes us a step closer to this reality.

This 370kg liquid Xenon experiment apparatus simulates neutron collisions using photons and xenon atoms

Exotic WIMPs, Axions and strange Quarks

By NASA's estimates only 4 per cent of the universe is made up of ordinary matter such as stars, planets and galaxies. The remaining 96 per cent is just a vast empty void thought to be made up of "dark matter" or "dark energy". Scientists have no physical evidence to determine what's out there, besides their mere observation of the gravitational influence of the nothingness. They believe that all those extra bits, remainders and constants that are left over in their mathematical models of the universe actually do exist – far away in the distant void of the empty 96 per cent.

To explain this cosmic absence, scientists the world over have turned to the idea of exotic matter, particles that behave contradictorily to the laws of physics.

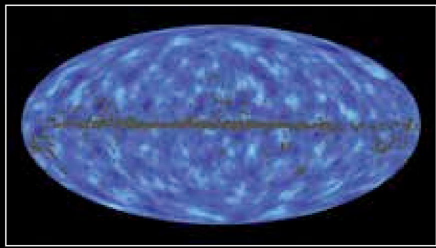
In October 2013, scientists at the Large Underground Xenon (LUX) project in South Dakota (USA) revealed that they were looking for a set of exotic particles known as 'weakly interacting massive particles' (WIMPs). They believe that the properties of WIMPs fit into the joint model which proves the presence of dark matter in the universe. But in the absence of any proof, the only way to make sense of their knowledge is to completely rewrite the laws of gravity under Einstein's general theory of relativity. This, of course, isn't an easy task and so far, has only led to failure.

WIMPs will not only help explain dark matter but will also fit nicely with the complex requirements of quantum theory such as supersymmetry and string theory. Researchers hope for even the slightest variation of WIMP type exotic particles as an encouraging sign since it would corroborate their hypothesis. Unfortunately, the quest for this particle has been on since the 1980s and has proven completely unsuccessful, despite LUX having super-sensitive equipment at its disposal.

People believe that if WIMPs aren't discovered over the next ten years, the odds of them being a real possibility are negligible. In order to explore alternatives, other scientists are already looking for another variety of exotic particles called 'axions'. These less massive particles are far harder to find as compared to WIMPs but with over a dozen active experiments active today, hope lives. Scientists believe that the mathematical modelling behind Axions is solid in its theoretical framework and could just as easily explain dark matter.

The University of Washington's Axion Dark Matter eXperiment (ADMX) has been in hot pursuit of an axion since 1995 but still hasn't found conclusive results. The experiment was recalibrated to a more sensitive iteration recently and is expected to find something in the next three years or finally set aside the idea of axions once and for all.

As the desperate times of null results weigh down scientists worldwide, another exotic particle theory is showing some



Such a vast universe and so little matter. Perhaps we're meant for more than just a physical existence?

promise – 'strange quark' matter. These denser versions of strange quarks are theorised to be formed inside massive neutron stars and fulfill the required criteria for proving dark matter presence in the universe. However, it's hard to detect these by nearly undetectable particles sitting here on Earth when they're only theorised to be found in neutron stars.

But there are worse outcomes than not finding strange quarks, axions and WIMPs; and that possibility is that dark matter is made of something that can't be found. That real astronomical exotic particles are only visible through their gravitational effects and nothing else. If this is true, then scientists have no chance of ever detecting them. This would be the ultimate failure for science: to have created a theory that can't be tested or proved.

It's not over yet

The universe is governed by invisible laws, which align all matter into patterns of cosmic beauty. Humans in turn, by their very nature, yearn to unravel these laws and have looked across the cosmos to seek meaning. These laws have proven elusive and mysterious throwing humanity deeper into uncertainty and doubt. But it is this very mystery that has motivated countless experts across history to uncover that which was once unimaginable.

The quest to find exotic particles and dark matter may seem fruitless now, but with CERN's LHC as the last bastion of hope, there's reason to still be hopeful. Even if that were to disappoint, be rest assured that we haven't surpassed the limits of human imagination and innovation. Everything is something and nothing is everything, and humans are trying to figure it all out one particle at a time. d